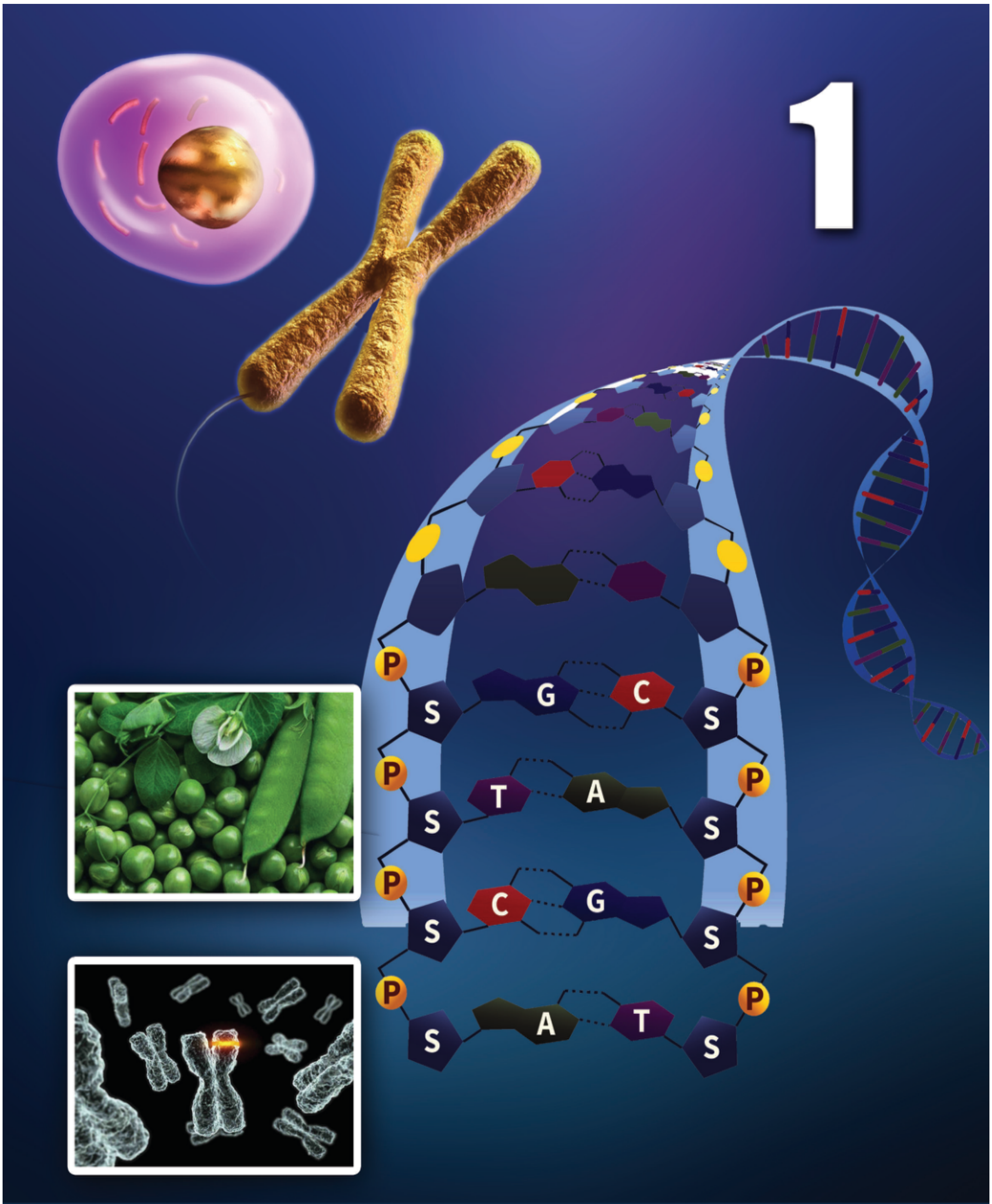


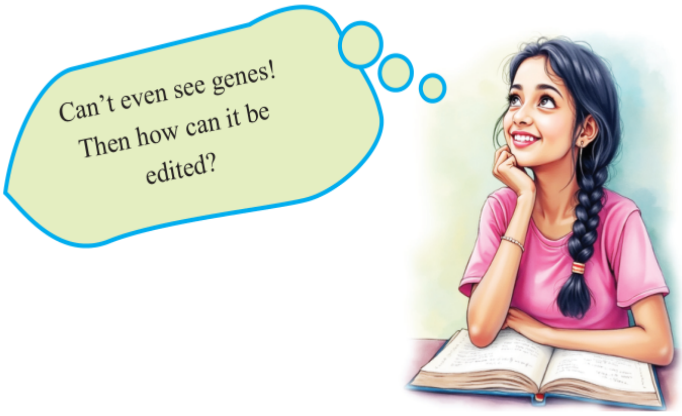
Genetics of Life

The Nobel Prize in Chemistry for developing methodology of Gene Editing

Jennifer A Doudna   Emmanuelle Charpentier

The Nobel Prize of 2020 in Chemistry was shared by Emmanuelle Charpentier and Jennifer A Doudna for their contributions in the field of gene editing. The award is for the discovery of a technology called CRISPR-Cas 9, a gene editing process which can bring desirable changes in the genes in DNA. This discovery is expected to make revolutionary advances in genetic disease therapy and treatment of cancer. It can also be used to develop crops that are resistant to pests and diseases.

You have read the extract related to the technology which can bring revolutionary changes in the treatment of diseases and other fields.



Didn't you too have this doubt?

A deeper understanding of DNA (Deoxyribonucleic acid) and genes paved the way for gene editing.

Let us understand the location and ultra structure of DNA and gene.

Observe illustration 1.1 and make notes on the location of DNA.

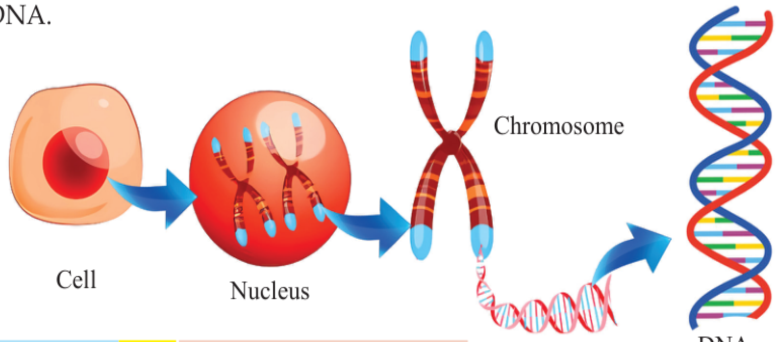


Illustration 1.1 The Location of DNA

You have understood the location of DNA. Discovery of the structure of the nucleic acid DNA was one of the greatest leaps in the field of biological studies.

Structure of DNA

In 1953, James Watson along with Francis Crick had presented the double helical model of DNA. They proposed the structure of DNA based on the X-ray diffraction studies conducted by Rosalind Franklin and Maurice Wilkins. The crucial information that led to this discovery was obtained from the famous 'Photo 51', an X-ray diffraction image of DNA taken by Rosalind Franklin. Rosalind Franklin passed away at the age of 37 in 1958. James Watson, Francis Crick and Maurice Wilkins were awarded the Nobel Prize in Medicine in 1962 for their contributions on the discovery of the double helix model of DNA.

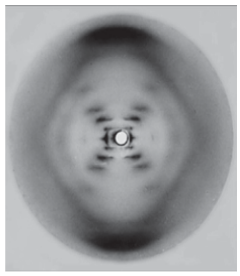


Photo 51

Rosalind Franklin 1920-1958 | Maurice Wilkins 1916-2004 | Francis Crick 1916-2004 | James Watson 1928

Analyse the illustration 1.2 to gain an understanding on the structure of DNA and complete the work sheet 1.1.

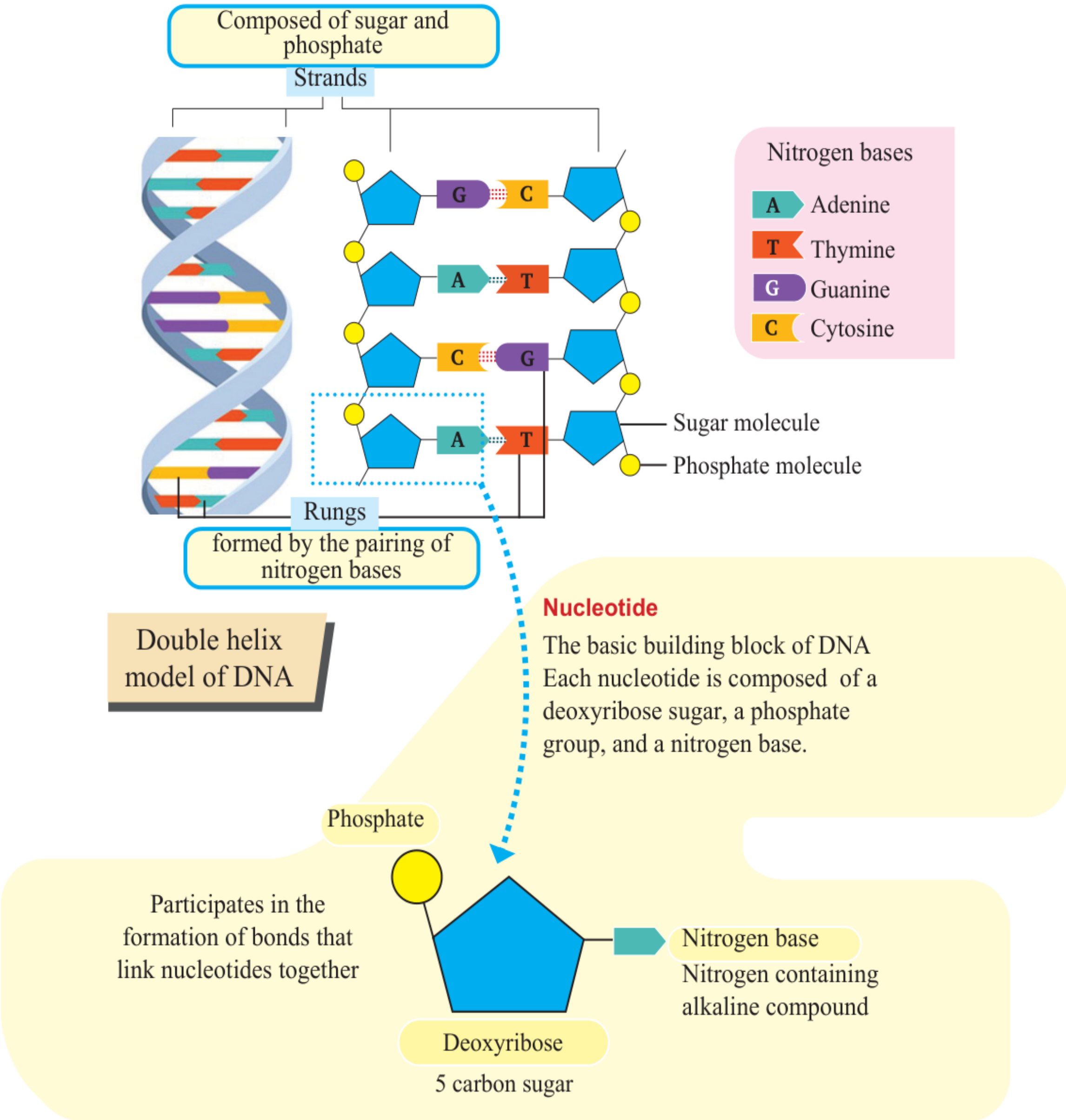


Illustration 1.2 Structure of DNA

Worksheet

1. Number of strands in DNA	
2. Molecules used to make strands	
3. Molecules used to make rungs	
4. Different types of nitrogen bases	
5. Formation of rungs	
6. Mode of nitrogen base pairing	
7. Molecules in a nucleotide	

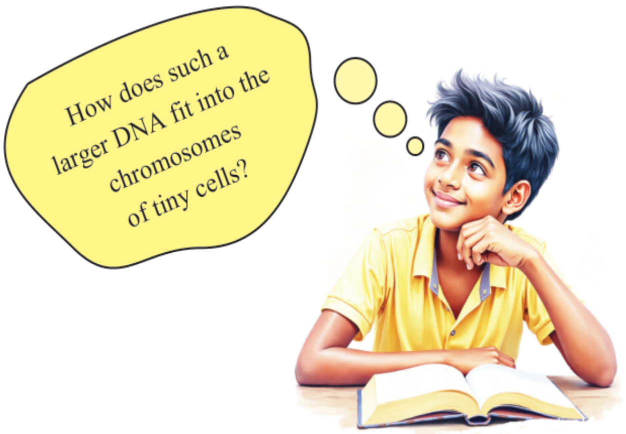
Worksheet 1.1 The structure of DNA

Prepare the double helix model of DNA by using locally available waste materials and display it in class.

How does normal sugar differ from a sugar molecule in DNA? Find out.

Size of DNA

The DNA in each chromosome is about 2 inches (5cm) long. If DNA from 46 chromosomes of a human cell joins together, it would be around 6 feet in length (2m). The human body is made up of trillions (one lakh crore) of cells. If the DNAs of all the cells joins together, it would be about 67 billion (one billion = 100 crore) miles in length. It is capable enough to wrap around the Earth over two million times!



Haven't you listened to the child's doubt?

Analyse the illustration 1.3 and description based on the indicators and find the answer to the child's doubt.